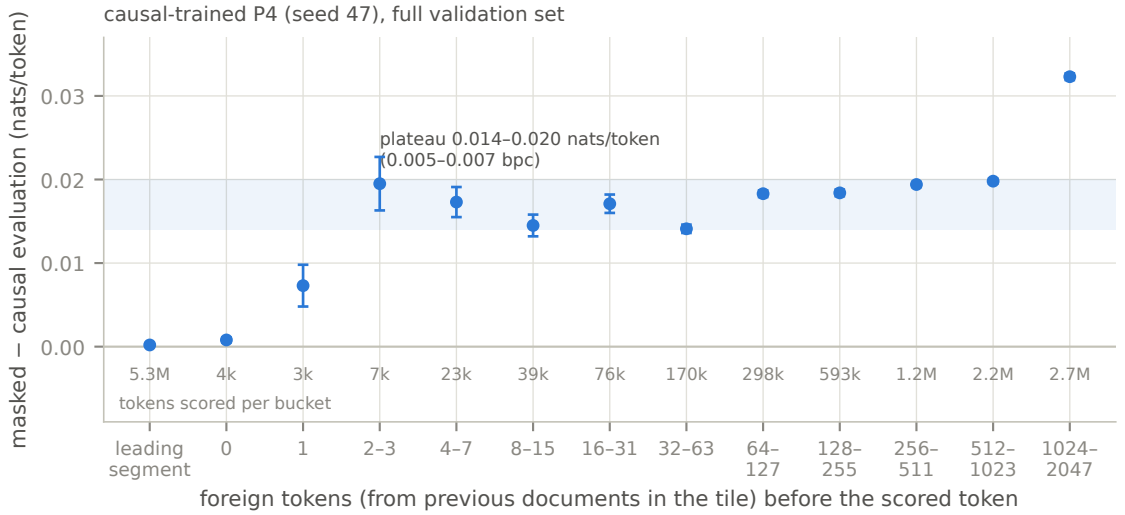
causal-trained P4 (seed 47), full validation set
masked − causal evaluation (nats/token)
0.03
0.02
0.01
0.00
plateau 0.014–0.020 nats/token
(0.005–0.007 bpc)
5.3M
4k
3k
7k
23k
39k
76k
170k
298k
593k
1.2M
2.2M
2.7M
tokens scored per bucket
leading segment
0
1
2–3
4–7
8–15
16–31
32–63
64–127
128–255
256–511
512–1023
1024–2047
foreign tokens (from previous documents in the tile) before the scored token